JIACHENG ZHU

4 Emerson Place, Apt 912, Boston, MA 02114
(+1) 734-882-9807 ⋄ zjc@csail.mit.edu ⋄ https://jiachengzhuml.github.io/ ⋄ Github

RESEARCH INTERESTS

Jiacheng Zhu's research focuses on developing generalizable and trustworthy foundation models and machine learning methods. Specifically, he works on fine-tuning, LLM efficiency and compression, mixture-of-experts (MoEs), mechanisms of foundation models, and their applications in healthcare and robotics. He extracts insights from Bayesian statistics, probabilistic modeling, and particularly optimal transport to investigate the underlying geometric structures within both the data and model parameters.

RESEARCH AND WORK EXPERIENCE

Postdoctoral Associate, *MIT CSAIL, Cambridge, MA, USA* — *Sep 2023 - Now*
Advisors: Justin Solomon, Marzyeh Ghassemi
Topics: Geometric-inspired foundational models & generalizable ML for healthcare

Research Assistant, *CMU, Pittsburgh, PA, USA* — *Sep 2018-Aug 2023*
Advisors: XuanLong Nguyen, Ding Zhao, Bo Li
Topic: Generalizable machine learning on heterogeneous domains with optimal transport

Research Scientist Intern, *Apple AI/ML Health AI, Seattle, WA, USA* — *May 2023-Aug 2023*
Mentors: Haraldur Hallgrímsson, Shirley Ren
Topic: Self-Supervised Learning (SSL) for time sequence data

Research Intern, *AT&T Labs, New York, NY, USA* — *May 2022-Aug 2022*
Mentors: Aritra Guha, Zhengyi Zhou, Cheryl Brooks
Topic: Statistical machine learning and algorithmic fairness

Research Scientist Intern, *Apple AI/ML Health AI, Seattle, WA, USA* — *May 2021-Oct 2021*
Mentors: Greg Darnell, Shirley Ren
Topic: Physiology-informed generalizable machine learning for time sequence data

Quantitative Research Intern, *Etiger Capital Partners, Shanghai, China* — *May 2017-Aug 2017*
Topic: Event-driven strategy based on probabilistic graphical models

EDUCATION

M.S. in Machine Learning, *Machine Learning Department* — *Dec 2019 - Jan 2022*
School of Computer Science, Carnegie Mellon University, PA, USA

Ph.D. Candidate, *Mechanical Engineering* — *Sept 2018 - Aug 2023*
College of Engineering, Carnegie Mellon University, PA, USA

Minor in Data Science, — *Sept 2015 - June 2017*
School of Computer Science, Fudan University, Shanghai, China

B.Eng. in Aerospace design and Engineering, — *Sept 2013 - June 2017*
School of Aeronautics and Astronautics, Fudan University, Shanghai, China

SELECTED PUBLICATIONS & PREPRINTS

Generalizable Machine Learning

1. **[JMLR] Functional Optimal Transport: Map Estimation and Domain Adaptation for Functional Data**,
Jiacheng Zhu*, Aritra Guha*, Dat Do*, Mengdi Xu, XuanLong Nguyen, Ding Zhao, *Journal of Machine Learning Research (JMLR), 2024.*

2. **[ICML FoMo workshop] Compress then Serve: Serving Thousands of LoRA Adapters with Little Overhead**,
Rickard Brüel Gabrielsson, **Jiacheng Zhu**, Onkar Bhardwa, Leshem Choshen,Kristjan Greenewald, Mikhail Yurochkin, Justin Solomon, under review, abridged at Efficient Systems for Foundation Models workshop on ICML 2024,

3. **[ICML 2024] Asymmetry in Low-Rank Adapters of Foundation Models**,
Jiacheng Zhu, Kristjan Greenewald, Kimia Nadjahi, Haitz Sáez de Ocáriz Borde, Rickard Brüel Gabrielsson, Leshem Choshen, Marzyeh Ghassemi, Mikhail Yurochkin, Justin Solomon, PMLR International Conference on Machine Learning (ICML), 2023.

4. **[CVPR 2024 Highlight] MMSum: A Dataset for Multimodal Summarization and Thumbnail Generation of Videos**,
Jielin Qiu, **Jiacheng Zhu**, William Han, Aditesh Kumar, Karthik Mittal, Claire Jin, Zhengyuan Yang, Linjie Li, Jianfeng Wang, Ding Zhao, Bo Li, Lijuan Wang. Conference on Computer Vision and Pattern Recognition (CVPR), 2024

5. **[NeurIPS 2023] Constraint-Conditioned Policy Optimization for Versatile Safe Reinforcement Learning**,
Yihang Yao, Zuxin Liu, Zhepeng Cen, **Jiacheng Zhu**, Wenhao Yu, Tingnan Zhang, Ding Zhao. Advances in Neural Information Processing Systems 36 (NeurIPS), 2023

6. **[ICML 2023] Interpolation for Robust Learning: Data Augmentation on Wasserstein Geodesics**,
Jiacheng Zhu, Jielin Qiu, Aritra Guha, Zhuolin Yang, Xuanlong Nguyen, Bo Li, Ding Zhao. PMLR International Conference on Machine Learning (ICML), 2023.

7. **[ACL 2023] Semantics-Consistent Cross-domain Summarization via Optimal Transport Alignment**,
Jielin Qiu, **Jiacheng Zhu**, Mengdi Xu, Frank Dernonocourt, Trung Bui, Zhaowen Wang, Bo Li, Ding Zhao, Hailin Jin, Annual Meeting of the Association for Computational Linguistics (ACL) 2023.

8. **[NeurIPS 2022] Curriculum Reinforcement Learning using Optimal Transport via Gradual Domain Adaptation**,
Peide Huang, Mengdi Xu, **Jiacheng Zhu**, Laixi Shi, Fei Fang, Ding Zhao, *Conference on Neural Information Processing Systems (NeurIPS)2022.*

9. **[NeurIPS 2020] Task-Agnostic Online Reinforcement Learning with an Infinite Mixture of Gaussian Processes**
Mengdi Xu, Wenhao Ding, **Jiacheng Zhu**, Zuxin Liu, Baiming Chen, Ding Zhao, *Conference on Neural Information Processing Systems (NeurIPS 2020).*

10. **[NeurIPS Workshop 2019] Recurrent Attentive Neural Process for Sequential Data**
Shenghao Qin*, **Jiacheng Zhu***, Jimmy Qin, Wenshuo Wang, Ding Zhao, *Conference on Neural Information Processing Systems (NeurIPS 2019) Workshop.*

Artificial Intelligence & Machine Learning for Health

7. **[ML4H 2024] Automated Cardiovascular Record Retrieval by Multimodal Learning between Electrocardiogram and Clinical Report**,
Jielin Qiu, **Jiacheng Zhu**, Shiqi Liu, William Han, Jingqi Zhang, Chaojing Duan, Michael A. Rosenberg, Emerson Liu, Douglas Weber, Ding Zhao, *the 3rd Machine Learning for Health Symposium PMLR (ML4H), 2023.*

8. **[ICASSP 2023] Cardiac Disease Diagnosis on Imbalanced Electrocardiography Data Through Optimal Transport Augmentation**,
Jielin Qiu*, **Jiacheng Zhu***, Mengdi Xu, Peide Huang, Michael Rosenberg, Douglas Weber,Emerson

Liu, Ding Zhao, *2023 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2023.*

9. [EACL 2023] **Transfer Knowledge from Natural Language to Electrocardiography: Can We Detect Cardiovascular Disease Through Language Models?**,
Jielin Qiu, William Han, **Jiacheng Zhu**, Mengdi Xu, Michael Rosenberg, Emerson Liu, Douglas Weber and Ding Zhao, *The 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL) 2023.*

10. [MLHC 2022] **GeoECG: Data Augmentation via Wasserstein Geodesic Perturbation for Robust Electrocardiogram Prediction**,
Jiacheng Zhu*, Jielin Qiu*, Zhuolin Yang, Douglas Weber, Michael Rosenberg, Emerson Liu, Bo Li, Ding Zhao, *PMLR Machine Learning for Healthcare (MLHC) 2022.*

11. [CHIL 2022] **PhysioMTL: Personalizing Physiological Patterns using Optimal Transport Multi-Task Regression**,
Jiacheng Zhu, Gregory Darnell, Agni Kumar, Ding Zhao, Bo Li, XuanLong Nguyen, Shirley You Ren, *PMLR Conference on Health, Inference, and Learning (CHIL) 2022.*

Artificial Intelligence & Machine Learning for Autonomy

11. [AISTATS 2024] **Pixel-wise Smoothing for Certified Robustness against Camera Motion Perturbations**,
Hanjiang Hu, Zuxin Liu, Linyi Li, **Jiacheng Zhu** and Ding Zhao, *2024 International Conference on Artificial Intelligence and Statistics (AISTATS), 2024.*

12. [RAL 2024] **Safety-Aware Causal Representation for Trustworthy Offline Reinforcement Learning in Autonomous Driving**,
Haohong Lin, Wenhao Ding, Zuxin Liu, Yaru Niu, **Jiacheng Zhu**, Yuming Niu, Ding Zhao, *IEEE Robotics and Automation Letters, 2024.*

13. [IROS 2024] **Seasondepth: Cross-season monocular depth prediction dataset and benchmark under multiple environments**,
Hanjiang Hu, Baoquan Yang, Zhijian Qiao, Shiqi Liu, **Jiacheng Zhu**, Zuxin Liu, Wenhao Ding, Ding Zhao, Hesheng Wang *IEEE Robotics and Automation Letters, 2024.*

14. [ITSC 2023] **Interactive Car-Following: Matters but NOT Always**,
Chengyuan Zhang, Rui Chen, **Jiacheng Zhu**, Wenshuo Wang, Changliu Liu, Lijun Sun *IEEE 26th International Conference on Intelligent Transportation Systems (ITSC), 2023.*

15. [IROS 2023] **Goats: Goal sampling adaptation for scooping with curriculum reinforcement learning**,
Yaru Niu, Shiyu Jin, Zeqing Zhang, **Jiacheng Zhu**, Ding Zhao, Liangjun Zhang, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.*

16. [CoRL 2022] **Robustness Certification of Visual Perception Models via Camera Motion Smoothing**,
Hanjiang Hu, Zuxin Liu, Linyi Li, **Jiacheng Zhu** Ding Zhao, *PMLR Conference on Robot Learning (CoRL) 2022.*

17. [IROS 2022] **Scalable Safety-Critical Policy Evaluation with Accelerated Rare Event Sampling**,
Mengdi Xu, Peide Huang, Fengpei Li, **Jiacheng Zhu**, Xuewei Qi, Kentaro Oguchi, Zhiyuan Huang, Henry Lam, and Ding Zhao, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022.*

18. [BuilSys 2022] **Re-vibe: vibration-based indoor person re-identification through cross-structure optimal transport**,

Yiwen Dong, **Jiacheng Zhu**, Hae Young Noh, *The 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys) 2022.*

19. **[TRC] Robust unsupervised learning of temporal dynamic vehicle-to-vehicle interactions**,
Aritra Guha, Rayleigh Lei, **Jiacheng Zhu**, XuanLong Nguyen, Ding Zhao, *Transportation research part C: emerging technologies.*

20. **[ICRA 2021] Context-Aware Safe Reinforcement Learning for Non-Stationary Environments**
Baiming Chen, Zuxin Liu, **Jiacheng Zhu**, Mengdi Xu, Wenhao Ding, Liang Li, Ding Zhao
The 2021 International Conference on Robotics and Automation (ICRA 2021).

21. **[T-ITS] Spatiotemporal learning of multivehicle interaction patterns in lane-change scenarios**
Chengyuan Zhang, **Jiacheng Zhu**, Wenshuo Wang, Junqiang Xi, *IEEE Transactions on Intelligent Transportation Systems, 2021.*

22. **[ITSC 2018] A Tempt to Unify Heterogeneous Driving Databases using Traffic Primitives**
Jiacheng Zhu, Wenshuo Wang, Ding Zhao, *The 21st IEEE International Conference on Intelligent Transportation Systems (ITSC) 2018.*

23. **[NME] A theoretical model for delayed hydride cracking velocity considering the temperature history and temperature gradients**
Jingyu Zhang, **Jiacheng Zhu**, Shurong Ding, etc., Nuclear Materials and Energy 16 (2018): 95-107.

24. **[SMiRT] Evaluation of the Effect of Temperature Gradient and Irradiation on Threshold Stress Intensity Factor in Delayed Hydride Cracking and Simulation on DHC Velocity for Zirconium Alloy Cladding Tubes**
Jiacheng Zhu, Jingyu Zhang, Shurong Ding, etc. oral report, proceedings with the 19th Conference on Structural Mechanics in Reactor Technology, Beijing, China, Oct 16-18, 2016

ACADEMIC SERVICES

Grant reviewer	2024 Tufts Small Grants to Advance Translational Science
Conference reviewer	International Conference on Machine Learning (ICML)
	Conference on Neural Information Processing Systems (NeurIPS)
	International Conference on Learning Representations (ICLR)
	Artificial Intelligence and Statistics Conference (AISTATS), (2023 top reviewer)
	Association for the Advancement of Artificial Intelligence (AAAI)
	Conference on Health, Inference, and Learning (CHIL)
	Machine Learning for Healthcare (MLHC)
Journal reviewer	Transactions on Machine Learning Research (TMLR)
	Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
	IEEE Robotics and Automation Letters (RA-L)
	IEEE Transactions on Intelligent Transportation Systems (T-ITS)
	IEEE Transactions on Intelligent Vehicles (T-IV)
Organizer	LLM-Merging competition at NeurIPS 2024 [llm-merging.github.io]
	CHIL 2024, Communication Chair
	ICRA 2022 SeasonDepth Challenge

TALKS & ACTIVITIES

Generalizing AI/ML for Physiological & Cardiovascular Health June 2023
Long talk Doctoral Symposium, Conference on Health, Inference, and Learning 2023, Cambridge, MA.

Generalizing AI/ML for Physiological Health with Heterogeneous Data April 2023
Long talk *AISOC Lab, CMU, USA, Hosted by Prof. Fei Fang, CMU*

Generalizing AI/ML for Autonomous Driving under Distribution Shift Oct 2022
Long talk *Wayve AI, London (remote), Hosted by Vijay Badrinarayanan,*

Robustify ECG Automatic diagnosis via distribution interpolation Sep 2022
Long talk *Allegheny Health Network, Pittsburgh*

GeoECG: Data augmentation for robust cardiovascular prediction Aug 2022
Spotlight presentation *Machine Learning for Healthcare, 2022, Durham, North Carolina*

PhysioMTL: Physiology-informed multi-task Learning April 2022
Spotlight presentation *Conference on Health, Inference, and Learning, 2022*

Functional Optimal Transport February 2022
Spotlight presentation *AAAI Workshop on Optimal Transport and Structured Data Modeling*

Demographic aware multitask learning for Heart Rate Variability August 2021
Intern presentation *Apple AI/ML, Remote*

Oxford Machine Learning Summer School July 2020
Attendee *OxML 2020, Oxford University (Online)*

Recurrent Attentive Neural Process for Sequential Data Dec 2019
Lightning talks *NeurIPS 2019,Learning with Rich Experience (LIRE) Workshop,Vancouver*

Artificial Intelligence for Data Discovery and Reuse (AIDR) 2019 May 2019
Long talk *Carnegie Mellon University, Pittsburgh*

HONORS

- Conference on Health, Inference, and Learning, Doctoral Symposium, 2023
- International Conference on Artificial Intelligence and Statistics (AISTATS) top reviewer, 2023
- Qualcomm Innovation Fellowship, North American, 2022
- Apple Scholars in AI/ML PhD fellowship nomination, 1 out of 3 at CMU College of Engineering, 2021
- Certification, Oxford Machine Learning Summer School, 2020 (Online)
- NeurIPS Student Travel Award, NeurIPS , 2019
- Rackham Travel Grant Fellowship, University of Michigan, Ann Arbor, 2018
- Junyuan Scholarship for Undergraduate Student, Fudan University, 2017
- Outstanding Undergraduate Thesis, Fudan University, 2016
- Xiexin Excellent Student Scholarship, Fudan University, 2016
- Junyuan Scholarship for Undergraduate Student, Fudan University, 2016
- 'GuangHua Innovation Prize' of Fudan University, Fudan University, 2015
- China Graduate Future Flight Vehicle Innovation Competition, First Prize, 2014

SELECTED COURSES

Statistical Machine Learning, Probabilistic Graphical Models, Convex Optimization, Reinforcement Learning, Advanced Deep Learning, Intro to Machine Learning, Computer Vision, Machine Learning with Large Datasets, Mobile Robotics: SLAM, Robot Kinematics & Dynamics, Database Management Systems, Data Structures and Algorithms

TECHNICAL AND PERSONAL SKILLS

Computer Languages	Python, JavaScript, C/C++, MATLAB
Softwares & Tools	TensorFlow, Pytorch, ROS, OpenRAVE, MySQL, OpenCV, CUDA, Solidworks, LaTeX
Languages	English, Mandarin, Shanghai Dialect